

150 MOST FREQUENTLY ASKED
QUESTIONS
ON QUANT INTERVIEWS

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To our beloved families

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Preface

The use of quantitative methods and programming skills in all areas of finance, from trading to risk management has grown tremendously in recent years, and accelerated through the financial crisis and with the advent of the big data era.

A core body of knowledge is required for successfully interviewing for a quant type position. The challenge lies in the fact that this knowledge encompasses finance, programming (in particular C++ programming), and several areas of mathematics (probability and stochastic calculus, numerical methods, linear algebra, and advanced calculus). Moreover, brainteasers are often asked to probe the ingenuity of candidates.

This book contains over 150 questions covering this core body of knowledge, without which it is not possible to advance to a final interview round. These questions are not only frequently, but also currently, asked on interviews for quantitative positions, and cover a vast spectrum, from C++ and data structures, to finance, brainteasers, and stochastic calculus.

The answers to all of these questions are included in the book. These answers are written in the same very practical vein that was used to select the questions: they are complete, but straight to the point – as they would be given in an interview.

Topics:

- Mathematics, calculus, differential equations.
- Covariance and correlation matrices. Linear algebra.
- Financial instruments: options, bonds, swaps, forwards, futures.
- C++, algorithms, data structures.
- Monte Carlo simulations. Numerical methods.
- Probability. Stochastic calculus.
- Brainteasers.

The authors are faculty members of the Baruch College Financial Engineering Masters Program, and have over 20 years of experience educating students who were very successful interviewing for quantitative positions. As such, the authors had the privilege to interact with generations of exceptional students, whose contributions as alumni to the continued success of our students has been tremendous. This book is a tribute to our special Baruch MFE community.

This is the first book in the *Pocket Book Guides for Quant Interviews* Series, to be followed by books on advanced probability and stochastic calculus questions and on challenging brainteasers asked in quant interviews.

New York, 2013

Acknowledgments

As professors in the Financial Engineering Masters Program at Baruch College, it has been a privilege to have the opportunity to contribute to the early career success of so many talented students and alumni. The strong community that developed around the Baruch MFE program is the reward educators truly dream of, and, for us, a reality that is an inspiration. This book would have not been possible without our Baruch MFE community – we are grateful to everyone who is part of it.

Working alongside and learning from our colleagues from the mathematics department and from the financial industry created the perfect context for writing this book; we are thankful to all of them.

Alejandro Cañete is owed a special thank you for his wonderful help on the programming questions, which are much improved due to him. The art for the book cover is, as always, due to the professional help of our alum and friend Max Rumyantsev. Several students spearheaded the proofreading effort, and their help was greatly appreciated: Yu Gan, Jun Hao, Alireza Kashef, Yi Bill Lu, Svetlana Rafailova, Fubo Shi, Yujia Helen Sun, Jun Charlotte Wang, Peng Wu, Yanzhu Wendy Wu, Yongyi Ivan Ye, He Hillary Zhao, and, last but not least, Wenyi Zhou.

It is the tremendous support and the understanding of our families that made this book possible, and we are

forever in debt to them. Thank you, thank you, and thank you!

This book is dedicated to our wives and to our children, from all our hearts.

Dan Stefanica

Radoš Radoičić

Tai-Ho Wang

New York, 2013

Chapter 1

First Look: Ten Questions.

1. Put options with strikes 30 and 20 on the same underlying asset and with the same maturity are trading for \$6 and \$4, respectively. Can you find an arbitrage?
2. The number 2^{29} has 9 digits, all different. Without computing 2^{29} , find the missing digit.
3. Let W_t be a Wiener process, and let

$$X_t = \int_0^t W_\tau d\tau.$$

What is the distribution of X_t ? Is X_t a martingale?

4. Alice and Bob stand at opposite ends of a straight line segment. Bob sends 50 ants towards Alice, one after another. Alice sends 20 ants towards Bob. All ants travel along the straight line segment. Whenever two ants collide, they simply bounce back and start traveling in the opposite direction. How many

ants reach Bob and how many ants reach Alice?
How many ant collisions take place?

5. Find all the values of ρ such that

$$\begin{pmatrix} 1 & 0.6 & -0.3 \\ 0.6 & 1 & \rho \\ -0.3 & \rho & 1 \end{pmatrix}$$

is a correlation matrix.

6. How many independent random variables uniformly distributed on $[0, 1]$ should you generate to ensure that there is at least one between 0.70 and 0.72 with probability 95%?
7. Show that the probability density function of the standard normal integrates to 1.
8. Assume the Earth is perfectly spherical and you are standing somewhere on its surface. You travel exactly 1 mile south, then 1 mile east, then 1 mile north. Surprisingly, you find yourself back at the starting point. If you are not at the North Pole, where can you possibly be?!
9. Solve the Ornstein-Uhlenbeck SDE

$$dr_t = \lambda(\theta - r_t)dt + \sigma dW_t,$$

which is used, e.g., in the Vasicek model for interest rates.

10. Write a C++ function that computes the n -th Fibonacci number.

Solutions

Question 1. Put options with strikes 30 and 20 on the same underlying asset and with the same maturity are trading for \$6 and \$4, respectively. Can you find an arbitrage?

Answer: Since the value of a put option with strike 0 is \$0, we in fact know the prices of put options with three different strikes, i.e.,

$$P(30) = 6; \quad P(20) = 4; \quad P(0) = 0,$$

where $P(K)$ denotes the value of a put option with strike K .

In the plane $(K, P(K))$, these option values correspond to the points $(30, 6)$, $(20, 4)$, and $(0, 0)$, which are on the line $P(K) = \frac{2}{3}K$.

This contradicts the fact that put options are strictly convex functions of strike price, and creates an arbitrage opportunity.

The arbitrage comes from the fact that the put with strike 20 is overpriced. Using a "buy low, sell high" strategy, we could buy (i.e., go long) $\frac{2}{3}$ put options with strike 30, and sell (i.e., go short) 1 put option with strike 20. To avoid fractions, we set up the following portfolio:

- long 2 puts with strike 30;
- short 3 puts with strike 20.

This portfolio is set up at no initial cost, since the cash flow generated by selling 3 puts with strike 20 and buying 2 puts with strike 30 is \$0:

$$3 \cdot \$4 - 2 \cdot \$6 = \$0.$$

At the maturity T of the options, the value of the portfolio is

$$V(T) = 2 \max(30 - S(T), 0) - 3 \max(20 - S(T), 0).$$

Note that $V(T)$ is nonnegative for any value $S(T)$ of the underlying asset:

If $S(T) \geq 30$, then both put options expire worthless, and $V(T) = 0$.

If $20 \leq S(T) < 30$, then

$$V(T) = 2(30 - S(T)) > 0.$$

If $0 < S(T) < 20$, then

$$\begin{aligned} V(T) &= 2(30 - S(T)) - 3(20 - S(T)) \\ &= S(T) \\ &> 0. \end{aligned}$$

In other words, we took advantage of the existing arbitrage opportunity by setting up, at no initial cost, a portfolio with nonnegative payoff at T regardless of the price $S(T)$ of the underlying asset, and with a strictly positive payoff if $0 < S(T) < 30$. $\square$

Question 2. The number 2^{29} has 9 digits, all different. Without computing 2^{29} , find the missing digit.

Answer: For any positive integer n , denote by $D(n)$ the sum of the digits of n . Recall that the difference between a number and the sum of its digits is divisible by 9, i.e.,

$$9 \mid n - D(n);$$

see the footnote below¹ for details.

¹If the digits of n are $a_k, a_{k-1}, \dots, a_1, a_0$ (from left to right), then

$$\begin{aligned} n &= a_k \cdot 10^k + a_{k-1} \cdot 10^{k-1} + \dots + a_1 \cdot 10 + a_0; \\ D(n) &= a_k + a_{k-1} + \dots + a_1 + a_0. \end{aligned}$$

Hence,

$$n - D(n) = \sum_{i=0}^k a_i \cdot (10^i - 1).$$

Since $10^i - 1$ is an i -digit number with all digits equal to 9, it follows that $9 \mid 10^i - 1$, for all $i = 1 : k$, and therefore $9 \mid n - D(n)$.

Thus, for $n = 2^{29}$, it follows that

$$9 \mid 2^{29} - D(2^{29}). \quad (1.1)$$

We are given that 2^{29} has 9 digits, and that all 9 digits are different. Denote by x the missing digit. Then,

$$D(2^{29}) = \left(\sum_{j=0}^9 j \right) - x = 45 - x. \quad (1.2)$$

From (1.1) and (1.2), it follows that

$$9 \mid 2^{29} - (45 - x). \quad (1.3)$$

Note that

$$\begin{aligned} 2^{29} &= 2^5 \cdot (2^6)^4 = 2^5 \cdot 64^4 \\ &= 2^5 \cdot (63 + 1)^4 \\ &= 2^5 \cdot (63 \cdot k + 1) \\ &= 2^5 \cdot 63 \cdot k + 2^5, \end{aligned} \quad (1.4)$$

where k is a positive integer.²

From (1.4), we find that

$$2^{29} - 2^5 = 63 \cdot 2^5 \cdot k,$$

and therefore

$$9 \mid 2^{29} - 2^5. \quad (1.5)$$

From (1.3) and (1.5), it follows that

$$\begin{aligned} 9 \mid (2^{29} - 2^5) - (2^{29} - (45 - x)) \\ = (45 - x) - 2^5 \\ = 13 - x. \end{aligned}$$

²It is easy to see that

$$\begin{aligned} (63 + 1)^4 &= 63^4 + 4 \cdot 63^3 + 6 \cdot 63^2 + 4 \cdot 63 + 1 \\ &= 63 \cdot (63^3 + 4 \cdot 63^2 + 6 \cdot 63 + 4) + 1 \\ &= 63 \cdot k + 1, \end{aligned}$$

where $k = 63^3 + 4 \cdot 63^2 + 6 \cdot 63 + 4$.

Since $9 \mid 13 - x$ and x is a digit, we conclude that $x = 4$. In other words, we identified that x , the missing digit from 2^{29} , must be 4.

Indeed, $2^{29} = 536\,870\,912$, i.e., 2^{29} has 9 digits, all different, and 4 is not a digit of 2^{29} . $\square$

Question 3. Let W_t be a Wiener process, and let

$$X_t = \int_0^t W_\tau d\tau. \quad (1.6)$$

What is the distribution of X_t ? Is X_t a martingale?

Answer: Note that we can rewrite (1.6) in differential form as

$$dX_t = W_t dt = W_t dt + 0 dW_t.$$

Then, X_t is a diffusion process with only drift part W_t , and therefore X_t is not a martingale.

We use integration by parts to find the distribution of X_t ; a different solution can be found in Section 3.6.

By applying integration by parts, we obtain that

$$\begin{aligned} X_t &= \int_0^t W_\tau d\tau \\ &= tW_t - \int_0^t \tau dW_\tau \\ &= t \int_0^t dW_\tau - \int_0^t \tau dW_\tau \\ &= \int_0^t (t - \tau) dW_\tau. \end{aligned}$$

Recall that, if $f(t)$ is a deterministic square integrable function, then the stochastic integral $\int_0^t f(\tau) dW_\tau$ is a normal random variable of mean 0 and variance $\int_0^t |f(\tau)|^2 d\tau$, i.e.,

$$\int_0^t f(\tau) dW_\tau \sim N\left(0, \int_0^t |f(\tau)|^2 d\tau\right).$$

Thus,

$$\begin{aligned} X_t &= \int_0^t (t - \tau) dW_\tau \\ &\sim N\left(0, \int_0^t (t - \tau)^2 d\tau\right) \\ &= N\left(0, \frac{t^3}{3}\right). \end{aligned}$$

We conclude that X_t is a normal random variable of mean 0 and variance $\frac{t^3}{3}$. $\square$

Question 4. Alice and Bob stand at opposite ends of a straight line segment. Bob sends 50 ants towards Alice, one after another. Alice sends 20 ants towards Bob. All ants travel along the straight line segment. Whenever two ants collide, they simply bounce back and start traveling in the opposite direction. How many ants reach Bob and how many ants reach Alice? How many ant collisions take place?

Answer: Imagine that when two ants meet, they switch identities. Hence, even after a collision, two ants are traveling in two opposite directions. It follows that 20 ants reach Bob, while 50 ants reach Alice.

To calculate the number of ant collisions, imagine that each ant carries a message. In other words, Bob sends 50 messages to Alice, one message per ant. Similarly, Alice sends 20 messages to Bob, one message per ant. Furthermore, imagine that the two ants swap messages when they collide. Then a message always makes forward progress. Each of Alice's messages goes through 50 ant collisions. Each of Bob's messages goes through 20 ant collisions. The total number of collisions is 50 times 20, which is 1000 collisions. $\square$

Question 5. Find all the values of ρ such that

$$\begin{pmatrix} 1 & 0.6 & -0.3 \\ 0.6 & 1 & \rho \\ -0.3 & \rho & 1 \end{pmatrix}$$

is a correlation matrix.

Answer: A symmetric matrix with diagonal entries equal to 1 is a correlation matrix if and only if the matrix is symmetric positive semidefinite. Thus, we need to find all the values of ρ such that the matrix

$$\Omega = \begin{pmatrix} 1 & 0.6 & -0.3 \\ 0.6 & 1 & \rho \\ -0.3 & \rho & 1 \end{pmatrix} \quad (1.7)$$

is symmetric positive semidefinite.

We give a short solution using Sylvester's criterion. Two more solutions, one using the Cholesky decomposition, and another one based on the definition of symmetric positive semidefinite matrices will be given in Section 3.2.

Recall from Sylvester's criterion that a matrix is symmetric positive semidefinite if and only if all its principal minors are greater than or equal to 0. Also, recall that the principal minors of a matrix are the determinants of all the square matrices obtained by eliminating the same rows and columns from the matrix. In particular, the matrix Ω from (1.7) has the following principal minors:

$$\det(1) = 1; \quad \det(1) = 1; \quad \det(1) = 1;$$

$$\det \begin{pmatrix} 1 & 0.6 \\ 0.6 & 1 \end{pmatrix} = 0.64;$$

$$\det \begin{pmatrix} 1 & -0.3 \\ -0.3 & 1 \end{pmatrix} = 0.91;$$

$$\det \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} = 1 - \rho^2;$$

$$\begin{aligned}\det(\Omega) &= 1 - 0.36\rho - 0.09 - 0.36 - \rho^2 \\ &= 0.55 - 0.36\rho - \rho^2.\end{aligned}$$

Thus, it follows from Sylvester's criterion that Ω is a symmetric positive semidefinite matrix if and only if

$$\begin{aligned}1 - \rho^2 &\geq 0; \\ 0.55 - 0.36\rho - \rho^2 &\geq 0,\end{aligned}$$

which is equivalent to $-1 \leq \rho \leq 1$ and

$$\rho^2 + 0.36\rho - 0.55 \leq 0. \quad (1.8)$$

Since the roots of the quadratic equation corresponding to (1.8) are -0.9432 and 0.5832 , we conclude that the matrix Ω is symmetric positive semidefinite, and therefore a correlation matrix, if and only if

$$-0.9432 \leq \rho \leq 0.5832. \quad \square \quad (1.9)$$

Question 6. How many independent random variables uniformly distributed on $[0, 1]$ should you generate to ensure that there is at least one between 0.70 and 0.72 with probability 95% ?

Answer: Denote by N the smallest number of random variables you should generate such that

$$P(\text{at least one r.v. in } [0.70, 0.72]) \geq 0.95. \quad (1.10)$$

The probability that a random variable uniformly distributed on $[0, 1]$ is not in the interval $[0.70, 0.72]$ is 0.98 . Thus, the probability that none of the N independent variables are in $[0.70, 0.72]$ is 0.98^N , i.e.,

$$P(\text{no r.v. in } [0.70, 0.72]) = 0.98^N.$$

Note that

$$\begin{aligned}
 & P(\text{at least one r.v. in } [0.70, 0.72]) \\
 &= 1 - P(\text{no r.v. in } [0.70, 0.72]) \\
 &= 1 - (0.98)^N. \qquad (1.11)
 \end{aligned}$$

From (1.10) and (1.11), we find that N is the smallest integer such that

$$1 - (0.98)^N \geq 0.95,$$

which is equivalent to

$$\begin{aligned}
 & (0.98)^N \leq 0.05 \\
 \iff & N \ln(0.98) \leq \ln(0.05) \\
 \iff & N \geq \frac{\ln(0.05)}{\ln(0.98)} \approx 148.28 \\
 \iff & N = 149.
 \end{aligned}$$

We conclude that at least 149 uniform random variables on $[0, 1]$ must be generated in order to have 95% confidence that at least one of the random variables is between 0.70 and 0.72. $\square$

Question 7. Show that the probability density function of the standard normal integrates to 1.

Answer: The probability density function of the standard normal variable is $\frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}$. We want to show that

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} dt = 1,$$

which, using the substitution $t = \sqrt{2}x$, can be written as

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}. \qquad (1.12)$$

We prove (1.12) by using polar coordinates. Since x is just an integrating variable, we can also write the integral I in terms of another integrating variable, denoted by y , as $I = \int_{-\infty}^{\infty} e^{-y^2} dy$.

Then,³

$$\begin{aligned} I^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \cdot \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-x^2} e^{-y^2} dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy. \end{aligned} \quad (1.13)$$

We use the polar coordinates transformation $x = r \cos \theta$ and $y = r \sin \theta$, with $r \in [0, \infty)$ and $\theta \in [0, 2\pi)$, to evaluate the last integral. Since $dx dy = r d\theta dr$, we obtain that

$$\begin{aligned} I^2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \\ &= \int_0^{\infty} \int_0^{2\pi} r e^{-(r^2 \cos^2 \theta + r^2 \sin^2 \theta)} d\theta dr \\ &= \int_0^{\infty} \int_0^{2\pi} r e^{-r^2} d\theta dr \\ &= \int_0^{\infty} 2\pi r e^{-r^2} dr \\ &= 2\pi \lim_{t \rightarrow \infty} \int_0^t r e^{-r^2} dr \\ &= 2\pi \lim_{t \rightarrow \infty} \left(-\frac{1}{2} e^{-r^2} \right) \Big|_0^t \\ &= \pi; \end{aligned} \quad (1.14)$$

note that (1.14) follows from the equality $\cos^2 \theta + \sin^2 \theta = 1$ for any real number θ .

³Note that Fubini's theorem is needed for a rigorous derivation of the equality (1.13); this technical step is rarely required by the interviewer.

Since $I > 0$, $I = \sqrt{\pi}$, which is what we wanted to prove; see (1.12). $\square$

Question 8. Assume the Earth is perfectly spherical and you are standing somewhere on its surface. You travel exactly 1 mile south, then 1 mile east, then 1 mile north. Surprisingly, you find yourself back at the starting point. If you are not at the North Pole, where can you possibly be?!

Answer: There are infinitely many locations, aside from the North Pole, that have this property.

Somewhere near the South Pole, there is a latitude that has a circumference of one mile. In other words, if you are at this latitude and start walking east (or west), in one mile you will be back exactly where you started from. If you instead start at some point one mile north of this latitude, your journey will take you one mile south to this special latitude, then one mile east “around the globe” and finally one mile north right back to wherever you started from. Moreover, there are infinitely many points on the Earth that are one mile north of this special latitude, where you could start your journey and eventually end up exactly where you started.

We are still not finished! There are infinitely many special latitudes as well; namely, you could start at any point one mile north of the latitude that has a circumference of $1/k$ miles, where k is a positive integer. Your journey will take you one mile south to this special latitude, then one mile east looping “around the globe” k times, and finally one mile north right back to where you started from. $\square$

Question 9. Solve the Ornstein-Uhlenbeck SDE

$$dr_t = \lambda(\theta - r_t)dt + \sigma dW_t, \quad (1.15)$$

which is used, e.g., in the Vasicek model for interest rates.

Answer: We can rewrite (1.15) as

$$dr_t + \lambda r_t dt = \lambda \theta dt + \sigma dW_t. \quad (1.16)$$

By multiplying (1.16) on both sides by the integrating factor $e^{\lambda t}$, we obtain that

$$e^{\lambda t} dr_t + \lambda e^{\lambda t} r_t dt = \lambda \theta e^{\lambda t} dt + \sigma e^{\lambda t} dW_t,$$

which is equivalent to

$$d(e^{\lambda t} r_t) = \lambda \theta e^{\lambda t} dt + \sigma e^{\lambda t} dW_t. \quad (1.17)$$

By integrating (1.17) from 0 to t , it follows that

$$\begin{aligned} e^{\lambda t} r_t - r_0 &= \lambda \theta \int_0^t e^{\lambda s} ds + \sigma \int_0^t e^{\lambda s} dW_s \\ &= \theta (e^{\lambda t} - 1) + \sigma \int_0^t e^{\lambda s} dW_s. \end{aligned}$$

By solving for r_t , we find that the solution to the Ornstein-Uhlenbeck SDE is

$$\begin{aligned} r_t &= e^{-\lambda t} r_0 + e^{-\lambda t} \theta (e^{\lambda t} - 1) + \sigma e^{-\lambda t} \int_0^t e^{\lambda s} dW_s \\ &= e^{-\lambda t} r_0 + \theta (1 - e^{-\lambda t}) + \sigma \int_0^t e^{-\lambda(t-s)} dW_s. \end{aligned}$$

Note that the process r_t is mean reverting to θ , regardless of the starting point r_0 . To see this, recall that the expected value of the stochastic integral $\int_0^t f(s) dW_s$ of a non-random function $f(s)$ is 0. Then,

$$E \left[\int_0^t e^{-\lambda(t-s)} dW_s \right] = 0,$$

and therefore

$$E[r_t] = e^{-\lambda t} r_0 + \theta (1 - e^{-\lambda t}).$$

Thus,

$$\lim_{t \rightarrow \infty} E[r_t] = \theta. \quad \square$$

Question 10. Write a C++ function that computes the n -th Fibonacci number.

Answer: The Fibonacci numbers $(F_n)_{n \geq 0}$ are given by the following recurrence:

$$F_{n+2} = F_{n+1} + F_n, \quad \forall n \geq 0,$$

with $F_0 = 0$ and $F_1 = 1$.

//recursive implementation

```
int fib(int n) {
    if (n == 0 || n == 1) return n;
    else {
        return fib(n-1) + fib(n-2);
    }
}
```

//iterative implementation

```
int fib(int n){
    if (n == 0 || n == 1) return n;
    int prev = 0, last = 1, temp;
    for (int i = 2; i <= n; ++i) {
        temp = last;
        last = prev + last;
        prev = temp;
    }
    return last;
}
```

//tail recursive implementation

```
int fib(int n, int last = 1, int prev = 0)
{
    if (n == 0) return prev;
```

```
    if (n == 1) return last;  
    return fib(n-1, last+prev, last);  
}
```

Chapter 2

Questions

1. Mathematics, calculus, differential equations.
2. Covariance and correlation matrices. Linear algebra.
3. Financial instruments: options, bonds, swaps, forwards, futures.
4. C++. Data structures.
5. Monte Carlo simulations. Numerical methods.
6. Probability. Stochastic calculus.
7. Brainteasers.

2.1 Mathematics, calculus, differential equations.

1. What is the value of i^i , where $i = \sqrt{-1}$?

2. Which number is larger, π^e or e^π ?

3. Show that

$$\frac{e^x + e^y}{2} \geq e^{\frac{x+y}{2}}, \quad \forall x, y \in \mathbb{R}.$$

4. Solve $x^6 = 64$.

5. What is the derivative of x^x ?

6. Calculate

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$$

7. Find x such that

$$x^{x^{x^{\dots}}} = 2.$$

8. Which of the following series converge:

$$\sum_{k=1}^{\infty} \frac{1}{k}; \quad \sum_{k=1}^{\infty} \frac{1}{k^2}; \quad \sum_{k=2}^{\infty} \frac{1}{k \ln(k)}?$$

9. Compute

$$\int \frac{1}{1+x^2} dx.$$

10. Compute

$$\int x \ln(x) dx \quad \text{and} \quad \int x e^x dx.$$

11. Compute

$$\int x^n \ln(x) dx.$$

12. Compute

$$\int (\ln(x))^n dx.$$

13. Solve the ODE

$$y'' - 4y' + 4y = 1.$$

14. Find $f(x)$ such that

$$f'(x) = f(x)(1 - f(x)).$$

15. Derive the Black-Scholes PDE.

2.2 Covariance and correlation matrices. Linear algebra.

1. Show that any covariance matrix is symmetric positive semidefinite. Show that the same is true for correlation matrices.
2. Find the correlation matrix of three random variables with covariance matrix

$$\Sigma_X = \begin{pmatrix} 1 & 0.36 & -1.44 \\ 0.36 & 4 & 0.80 \\ -1.44 & 0.80 & 9 \end{pmatrix}.$$

3. Assume that all the entries of an $n \times n$ correlation matrix which are not on the main diagonal are equal to ρ . Find upper and lower bounds on the possible values of ρ .
4. How many eigenvalues does an $n \times n$ matrix with real entries have? How many eigenvectors?

5. Let

$$A = \begin{pmatrix} 2 & -2 \\ -2 & 5 \end{pmatrix}.$$

- (i) Find a 2×2 matrix M such that $M^2 = A$;
- (ii) Find a 2×2 matrix M such that $A = MM^t$.

6. The 2×2 matrix A has eigenvalues 2 and -3 with corresponding eigenvectors $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$. If

$$v = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \text{ find } Av.$$

7. Let A and B be square matrices of the same size. Show that the traces of the matrices AB and BA are equal.

8. Can you find $n \times n$ matrices A and B such that

$$AB - BA = I_n,$$

where I_n is the identity matrix of size n ?

9. A probability matrix is a matrix with nonnegative entries such that the sum of the entries in each row of the matrix is equal to 1. Show that the product of two probability matrices is a probability matrix.

10. Find all the values of ρ such that

$$\begin{pmatrix} 1 & 0.6 & -0.3 \\ 0.6 & 1 & \rho \\ -0.3 & \rho & 1 \end{pmatrix}$$

is a correlation matrix.¹

¹A solution to this question was given in Chapter 1 using Sylvester's criterion; two different solutions will be given herein.

2.3 Financial instruments: options, bonds, swaps, forwards, futures.

1. The prices of three put options with strikes 40, 50, and 70, but otherwise identical, are \$10, \$20, and \$30, respectively. Is there an arbitrage opportunity present? If yes, how can you make a riskless profit?
2. The price of a stock is \$50. In three months, it will either be \$47 or \$52, with 50% probability. How much would you pay for an at-the-money put? Assume for simplicity that the stock pays no dividends and that interest rates are zero.
3. A stock worth \$50 today will be worth either \$60 or \$40 in three months, with equal probability. The value of a three months at-the-money put on this stock is \$4. Does the value of the three months ATM put increase or decrease, and by how much, if the probability of the stock going up to \$60 were 75% and the probability of the stock going down to \$40 were 25%?
4. What is risk-neutral pricing?
5. Describe briefly how you arrive at the Black-Scholes formula.
6. How much should a three months at-the-money put on an asset with spot price \$40 and volatility 30% be worth? Assume, for simplicity, that interest rates are zero and that the asset does not pay dividends.

7. If the price of a stock doubles in one day, by how much will the value of a call option on this stock change?
8. What are the smallest and largest values that Delta can take?
9. What is the Delta of an at-the-money call? What is the Delta of an at-the-money put?
10. What is the Put-Call parity? How do you prove it?
11. Show that the time value of a European call option is highest at-the-money.
12. What is implied volatility? What is a volatility smile? How about a volatility skew?
13. What is the Gamma of an option? Why is it preferable to have small Gamma? Why is the Gamma of plain vanilla options positive?
14. When are a European call and a European put worth the same? (The options are written on the same asset and have the same strike and maturity.) What is the intuition behind this result?
15. What is the two year volatility of an asset with 30% six months volatility?

16. How do you value an interest rate swap?
17. By how much will the price of a ten year zero coupon bond change if the yield increases by ten basis points?
18. A five year bond with 3.5 years duration is worth 102. What is the value of the bond if the yield decreases by fifty basis points?
19. What is a forward contract? What is the forward price?
20. What is the forward price for treasury futures contracts? What is the forward price for commodities futures contracts?
21. What is a Eurodollar futures contract?
22. What are the most important differences between forward contracts and futures contracts?
23. What is the ten-day 99% VaR of a portfolio with a five-day 98% VaR of \$10 million?

2.4 C++. Data structures.

1. How do you declare an array?
2. How do you get the address of a variable?
3. How do you declare an array of pointers?
4. How do you declare a const pointer, a pointer to a const and a const pointer to a const?
5. How do you declare a dynamic array?
6. What is the general form for a function signature?
7. How do you pass-by-reference?
8. How do you pass a read only argument by reference?
9. What are the important differences between using a pointer and a reference?
10. How do you set a default value for a parameter?
11. How do you create a template function?

12. How do you declare a pointer to a function?
13. How do you prevent the compiler from doing an implicit conversion with your class?
14. Describe all the uses of the keyword `static` in C++.
15. Can a static member function be `const`?
16. C++ constructors support the initialization of member data via an initializer list. When is this preferable to initialization inside the body of the constructor?
17. What is a copy constructor, and how can the default copy constructor cause problems when you have pointer data members?
18. What is the output of the following code:

```
#include <iostream>
using namespace std;
```

```
class A
{
public:
    int * ptr;
    ~A()
    {
        delete(ptr);
    }
};
```

```
void foo(A object_input)
{
    ;
}

int main()
{
    A aa;
    aa.ptr = new int(2);
    foo(aa);
    cout<<(*aa.ptr)<<endl;
    return 0;
}
```

19. How do you overload an operator?
20. What are smart pointers?
21. What is encapsulation?
22. What is a polymorphism?
23. What is inheritance?
24. What is a virtual function? What is a pure virtual function and when do you use it?
25. Why are virtual functions use for destructors? Can they be used for constructors?

26. Write a function that computes the factorial of a positive integer.
27. Write a function that takes an array and returns the subarray with the largest sum.
28. Write a function that returns the prime factors of a positive integer.
29. Write a function that takes a 64-bit integer and swaps the bits at indices i and j .
30. Write a function that reverses a single linked list.
31. Write a function that takes a string and returns true if its parenthesis are balanced.
32. Write a function that returns the height of an arbitrary binary tree.

2.5 Monte Carlo simulations. Numerical methods.

1. How would you compute π using Monte Carlo simulations? What is the standard deviation of this method?
2. What methods do you know for generating independent samples of the standard normal distribution?
3. How do you generate a geometric Brownian motion stock path using random numbers from a normal distribution?
4. How do you generate a sample of the standard normal distribution from 12 independent samples of the uniform distribution on $[0, 1]$?
5. What is the rate of convergence for Monte Carlo methods?
6. What variance reduction techniques do you know?
7. How do you generate samples of normal random variables with correlation ρ ?
8. What is the order of convergence of the Newton's method?

9. Which finite difference method corresponds to trinomial trees?

2.6 Probability. Stochastic calculus.

1. What is the exponential distribution? What are the mean and the variance of the exponential distribution?
2. If X and Y are independent exponential random variables with mean 6 and 8, respectively, what is the probability that Y is greater than X ?
3. What are the expected value and the variance of the Poisson distribution?
4. A point is chosen uniformly from the unit disk. What is the expected value of the distance between the point and the center of the disk?
5. Consider two random variables X and Y with mean 0 and variance 1, and with joint normal distribution. If $\text{cov}(X, Y) = \frac{1}{\sqrt{2}}$, what is the conditional probability $P(X > 0 | Y < 0)$?
6. If X and Y are lognormal random variables, is their product XY lognormally distributed?
7. Let X be a normal random variable with mean μ and variance σ^2 , and let Φ be the cumulative distribution function of the standard normal distribution. Find the expected value of $Y = \Phi(X)$.

8. What is the law of large numbers?
9. What is the central limit theorem?
10. What is a martingale? How is it related to option pricing?
11. Explain the assumption $(dW_t)^2 = dt$ used in the informal derivation of Itô's Lemma.
12. If W_t is a Wiener process, find $E[W_t W_s]$.
13. If W_t is a Wiener process, what is $\text{var}(W_t + W_s)$?
14. Let W_t be a Wiener process. Find

$$\int_0^t W_s dW_s \text{ and } E \left[\int_0^t W_s dW_s \right].$$

15. Find the distribution of the random variable

$$X = \int_0^1 W_t dW_t.$$

16. Let W_t be a Wiener process. Find the mean and the variance of

$$\int_0^t W_s^2 dW_s.$$

17. If W_t is a Wiener process, find the variance of

$$X = \int_0^1 \sqrt{t} e^{\frac{W_t^2}{8}} dW_t.$$

18. If W_t is a Wiener process, what is $E[e^{W_t}]$?

19. If W_t is a Wiener process, find the variance of

$$\int_0^t s dW_s.$$

20. Let W_t be a Wiener process, and let

$$X_t = \int_0^t W_\tau d\tau.$$

What is the distribution of X_t ? Is X_t a martingale?²

21. What is an Itô process?

22. What is Itô's lemma?

23. If W_t is a Wiener process, is the process $X_t = W_t^2$ a martingale?

²A solution to this question was given in Chapter 1 using integration by parts; a different solution will be given herein.

24. If W_t is a Wiener process, is the process

$$N_t = W_t^3 - 3tW_t$$

a martingale?

25. What is Girsanov's theorem?

26. What is the martingale representation theorem, and how is it related to option pricing and hedging?

27. Solve $dY_t = Y_t dW_t$, where W_t is a Wiener process.

28. Solve the following SDEs:

(i) $dY_t = \mu Y_t dt + \sigma Y_t dW_t$;

(ii) $dX_t = \mu dt + (aX_t + b)dW_t$.

29. What is the Heston model?

2.7 Brainteasers.

1. A flea is going between two points which are 100 inches apart by jumping (always in the same direction) either one inch or two inches at a time. How many different paths can the flea travel by?
2. I have a bag containing three pancakes: one golden on both sides, one burnt on both sides, and one golden on one side and burnt on the other. You shake the bag, draw a pancake at random, look at one side, and notice that it is golden. What is the probability that the other side is golden?
3. Alice and Bob are playing heads and tails, Alice tosses $n + 1$ coins, Bob tosses n coins. The coins are fair. What is the probability that Alice will have strictly more heads than Bob?
4. Alice is in a restaurant trying to decide between three desserts. How can she choose one of three desserts with equal probability with the help of a fair coin? What if the coin is biased and the bias is unknown?
5. What is the expected number of times you must flip a fair coin until it lands on head? What if the coin is biased and lands on head with probability p ?
6. What is the expected number of coin tosses of a fair coin in order to get two heads in a row? What if

the coin is biased with 25% probability of getting heads?

7. A fair coin is tossed n times. What is the probability that no two consecutive heads appear?
8. You have two identical Fabergé eggs, either of which would break if dropped from the top of a building with 100 floors. Your task is to determine the highest floor from which the an egg could be dropped without breaking. What is the minimum number of drops required to achieve this? You are allowed to break both eggs in the process.
9. An ant is in the corner of a $10 \times 10 \times 10$ room and wants to go to the opposite corner. What is the length of the shortest path the ant can take?
10. A $10 \times 10 \times 10$ cube is made of 1,000 unit cubes. How many unit cubes can you see on the outside?
11. Fox Mulder is imprisoned by aliens in a large circular field surrounded by a fence. Outside the fence is a vicious alien that can run four times as fast as Mulder, but is constrained to stay near the fence. If Mulder can contrive to get to an unguarded point on the fence, he can quickly scale the fence and escape. Can he get to a point on the fence ahead of the alien?
12. At your subway station, you notice that of the two trains running in opposite directions which are supposed to arrive with the same frequency, the train

going in one direction comes first 80% of the time, while the train going in the opposite direction comes first only 20% of the time. What do you think could be happening?

13. You start off with one amoeba. Every minute, this amoeba can either die, do nothing, split into two amoebas, or split into three amoebas; all these scenarios being equally likely to happen. All further amoebas behave the same way. What is the probability that the amoebas eventually die off?
14. Given a set X with n elements, choose two subsets A and B at random. What is the probability of A being a subset of B ?
15. Alice writes two distinct real numbers between 0 and 1 on two sheets of paper. Bob selects one of the sheets randomly to inspect it. He then has to declare whether the number he sees is the bigger or smaller of the two. Is there any way Bob can expect to be correct more than half the times Alice plays this game with him?
16. How many digits does the number 125^{100} have? You are not allowed to use values of $\log_{10} 2$ or $\log_{10} 5$.
17. For every subset of $\{1, 2, 3, \dots, 2013\}$, arrange the numbers in the increasing order and take the sum with alternating signs. The resulting integer is called

the weight of the subset.³ Find the sum of the weights of all the subsets of $\{1, 2, 3, \dots, 2013\}$.

18. Alice and Bob alternately choose one number from one of the following nine numbers: $1/16$, $1/8$, $1/4$, $1/2$, 1 , 2 , 4 , 8 , 16 , without replacement. Whoever gets three numbers that multiply to one wins the game. Alice starts first. What should her strategy be? Can she always win?
19. Mr. and Mrs. Jones invite four other couples over for a party. At the end of the party, Mr. Jones asks everyone else how many people they shook hands with, and finds that everyone gives a different answer. Of course, no one shook hands with his or her spouse and no one shook the same person's hand twice. How many people did Mrs. Jones shake hands with?
20. The New York Yankees and the San Francisco Giants are playing in the World Series (best of seven format). You would like to bet \$100 on the Yankees winning the World Series, but you can only place bets on individual games, and every time at even odds. How much should you bet on the first game?
21. We have two red, two green and two yellow balls. For each color, one ball is heavy and the other is light. All heavy balls weigh the same. All light balls weigh the same. How many weighings on a scale are necessary to identify the three heavy balls?

³For example, the weight of the subset $\{3\}$ is 3. The weight of the subset $\{2, 5, 8\}$ is $2 \cdot 5 \cdot 8 = 5$.

22. There is a row of 10 rooms and a treasure in one of them. Each night, a ghost moves the treasure to an adjacent room. You are trying to find the treasure, but can only check one room per day. How do you find it?
23. How many comparisons do you need to find the maximum in a set of n distinct numbers? How many comparisons do you need to find both the maximum and minimum in a set of n distinct numbers?
24. Given a cube, you can jump from one vertex to a neighboring vertex with equal probability. Assume you start from a certain vertex (does not matter which one). What is the expected number of jumps to reach the opposite vertex?
25. Select numbers uniformly distributed between 0 and 1, one after the other, as long as they keep decreasing; i.e. stop selecting when you obtain a number that is greater than the previous one you selected.
26. To organize a charity event that costs \$100K, an organization raises funds. Independent of each other, one donor after another donates some amount of money that is exponentially distributed with a mean of \$20K. The process is stopped as soon as \$100K or more has been collected. Find the distribution, mean, and variance of the number of donors needed until at least \$100K has been collected.

27. Consider a random walk starting at 1 and with equal probability of moving to the left or to the right by one unit, and stopping either at 0 or at 3.
- (i) What is the expected number of steps to do so?
 - (ii) What is the probability of the random walk ending at 3 rather than at 0?
28. A stick of length 1 drops and breaks at a random place uniformly distributed across the length. What is the expected length of the smaller part?
29. You are given a stick of unit length.
- (i) The stick drops and breaks at two places. What is the probability that the three pieces could form a triangle?
 - (ii) The stick drops and breaks at one place. Then the larger piece is taken and dropped again, breaking at one place. What is the probability that the three pieces could form a triangle?
30. Why is a manhole cover round?
31. When is the first time after 12 o'clock that the hour and minute hands of a clock meet again?
32. Three light switches are in one room, and they turn three light bulbs in another. How do you figure out which switch turns on which bulb in one shot?

Chapter 3

Solutions

3.1 Mathematics, calculus, differential equations.

Question 1. What is the value of i^i , where $i = \sqrt{-1}$?

Answer: Recall that $e^{i\theta} = \cos \theta + i \sin \theta$. Then,

$$i = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = e^{i\frac{\pi}{2}},$$

and therefore

$$i^i = \left(e^{i\frac{\pi}{2}}\right)^i = e^{(i\frac{\pi}{2})i} = e^{i^2\frac{\pi}{2}} = e^{-\frac{\pi}{2}},$$

since $i^2 = -1$. $\square$

Question 2. Which number is larger, π^e or e^π ?

Answer: We will show that $\pi^e < e^\pi$. By taking the natural logarithm, we find that

$$\begin{aligned}\pi^e < e^\pi &\iff \ln(\pi^e) < \ln(e^\pi) \iff e \ln(\pi) < \pi \\ &\iff \frac{\ln(\pi)}{\pi} < \frac{1}{e},\end{aligned}\tag{3.1}$$

which can be written as

$$\frac{\ln(\pi)}{\pi} < \frac{\ln(e)}{e}.$$

Let $f : (0, \infty) \rightarrow \mathbb{R}$ given by $f(x) = \frac{\ln(x)}{x}$. Then,

$$f'(x) = \frac{1 - \ln(x)}{x^2}.$$

Note that $f'(x) = 0$ has one solution, $x = e$. Also, $f'(x) > 0$ for $0 < x < e$, and $f'(x) < 0$ for $x > e$, and therefore $f(x)$ is increasing on the interval $(0, e)$ and is decreasing on the interval (e, ∞) .

Thus, the function $f(x) = \frac{\ln(x)}{x}$ has a global maximum point at $x = e$, i.e., $f(x) < f(e) = \frac{1}{e}$ for all $x > 0$ with $x \neq e$, and therefore

$$f(\pi) = \frac{\ln(\pi)}{\pi} < \frac{\ln(e)}{e} = \frac{1}{e},$$

which is equivalent to $\pi^e < e^\pi$; cf. (3.1). $\square$

Question 3. Show that

$$\frac{e^x + e^y}{2} \geq e^{\frac{x+y}{2}}, \quad \forall x, y \in \mathbb{R}. \quad (3.2)$$

Answer: Let $e^x = a$ and $e^y = b$. Note that $a, b > 0$, and that

$$e^{\frac{x+y}{2}} = \sqrt{e^{x+y}} = \sqrt{e^x \cdot e^y} = \sqrt{ab}.$$

Then, (3.2) can be written as

$$\begin{aligned} \frac{a+b}{2} \geq \sqrt{ab} &\iff a+b - 2\sqrt{ab} \geq 0 \\ &\iff (\sqrt{a} - \sqrt{b})^2 \geq 0, \end{aligned}$$

which is what we wanted to show. $\square$

Question 4. Solve $x^6 = 64$.

Answer: Recall that the six unit roots of $z^6 = 1$ are

$$\begin{aligned} z_k &= \exp\left(\frac{2k\pi i}{6}\right) = \exp\left(\frac{k\pi i}{3}\right) \\ &= \cos\left(\frac{k\pi}{3}\right) + i \sin\left(\frac{k\pi}{3}\right), \end{aligned} \quad (3.3)$$

for $k = 0 : 5$. Since $\sqrt[6]{64} = 2$, we obtain from (3.3) that the solutions of $x^6 = 64$ are

$$x_k = 2 \cos\left(\frac{k\pi}{3}\right) + 2i \sin\left(\frac{k\pi}{3}\right), \quad \forall k = 0 : 5. \quad \square$$

Question 5. What is the derivative of x^x ?

Answer: Note that

$$x^x = e^{\ln(x^x)} = e^{x \ln(x)}. \quad (3.4)$$

Using Chain Rule and (3.4), we find that

$$\begin{aligned} (x^x)' &= \left(e^{x \ln(x)}\right)' = e^{x \ln(x)} (x \ln(x))' \\ &= x^x (\ln(x) + 1). \quad \square \end{aligned}$$

Question 6. Calculate

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}} \quad (3.5)$$

Answer: Assume that the limit from (3.5) exists, and denote that limit by l . Then, $l = \sqrt{2+l}$, which can be written as

$$l^2 - l - 2 = (l-2)(l+1) = 0.$$

Since $l > 0$, we obtain that $l = 2$, i.e.,

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}} = 2. \quad (3.6)$$

Thus, proving (3.6) is equivalent to showing that the sequence $(x_n)_{n \geq 0}$ given by $x_0 = \sqrt{2}$ and

$$x_{n+1} = \sqrt{2 + x_n}, \quad \forall n \geq 0,$$

is convergent.

We can see by induction that the sequence $(x_n)_{n \geq 0}$ is bounded from above by 2, since $x_0 = \sqrt{2} < 2$, and, if we assume that $x_n < 2$, then

$$x_{n+1} = \sqrt{2 + x_n} < \sqrt{4} = 2.$$

Moreover, the sequence $(x_n)_{n \geq 0}$ is increasing, since

$$\begin{aligned} x_n < x_{n+1} &\iff x_n < \sqrt{2 + x_n} \\ &\iff x_n^2 - x_n - 2 < 0 \\ &\iff (x_n - 2)(x_n + 1) < 0, \end{aligned}$$

which holds true since $x_n > 0$, and since, as shown above, $x_n < 2$ for all $n \geq 0$.

Thus, the sequence $(x_n)_{n \geq 0}$ is convergent since it is increasing and bounded from above, which is what we needed to show in order to prove (3.6). $\square$

Question 7. Find x such that

$$x^{x^{x^{\dots}}} = 2. \quad (3.7)$$

Answer: If x exists such that (3.7) holds true, then

$$x^{x^{x^{\dots}}} = x^2 = 2,$$

and therefore the only possible solution to (3.7) is $x = \sqrt{2}$. We prove that $x = \sqrt{2}$ is, indeed, the solution to (3.7), by showing that

$$\sqrt{2}^{\sqrt{2}^{\sqrt{2}^{\cdot}}} = 2. \quad (3.8)$$

Consider the sequence $(x_n)_{n \geq 0}$ with $x_0 = \sqrt{2}$ and satisfying the following recursion:

$$x_{n+1} = \left(\sqrt{2}\right)^{x_n} = 2^{x_n/2}, \quad \forall n \geq 0. \quad (3.9)$$

We can see by induction that $(x_n)_{n \geq 0}$ is an increasing sequence, since $x_0 = \sqrt{2} < \sqrt{2}^{\sqrt{2}} = x_1$, and, if we assume that $x_{n-1} < x_n$, then

$$x_n = 2^{x_{n-1}/2} < 2^{x_n/2} = x_{n+1}.$$

Also by induction, we can prove that the sequence $(x_n)_{n \geq 0}$ is bounded from above by 2, since $x_0 = \sqrt{2} < 2$, and, if $x_n < 2$, then

$$x_{n+1} = 2^{x_n/2} < 2.$$

Thus, the sequence $(x_n)_{n \geq 0}$ is convergent since it is increasing and bounded from above.

Let $l = \lim_{n \rightarrow \infty} x_n$. From (3.9), we find that $l = 2^{l/2}$, which is equivalent to

$$l^{1/l} = 2^{1/2}. \quad (3.10)$$

The function $f : (0, \infty) \rightarrow (0, \infty)$ given by

$$\begin{aligned} f(t) &= t^{1/t} = \exp\left(\ln(t^{1/t})\right) \\ &= \exp\left(\frac{\ln(t)}{t}\right) \end{aligned}$$

is increasing for $t < e$ and decreasing for $t > e$, since

$$f'(t) = \frac{1 - \ln(t)}{t^2} \exp\left(\frac{\ln(t)}{t}\right)$$

and $f'(t) > 0$ for $t < e$ and $f'(t) < 0$ for $t > e$.

Thus, there are two values of l such that (3.10) is satisfied, i.e., such that $l^{1/l} = 2^{1/2}$, one value being equal to 2, and the other one greater than e . Since we showed that $x_n < 2$ for all $n \geq 0$, we conclude that $l = \lim_{n \rightarrow \infty} x_n = 2$, which is what we needed to show to complete the proof of (3.8). $\square$

Question 8. Which of the following series converge:

$$\sum_{k=1}^{\infty} \frac{1}{k}; \quad \sum_{k=1}^{\infty} \frac{1}{k^2}; \quad \sum_{k=2}^{\infty} \frac{1}{k \ln(k)}?$$

Answer: We show that

$$\sum_{k=1}^{\infty} \frac{1}{k^2} \quad \text{is convergent};$$

$$\sum_{k=1}^{\infty} \frac{1}{k} \quad \text{and} \quad \sum_{k=2}^{\infty} \frac{1}{k \ln(k)} \quad \text{are divergent.}$$

Since all the terms of the series $\sum_{k=1}^{\infty} \frac{1}{k^2}$ are positive, it is enough to show that the partial sums $\sum_{k=1}^n \frac{1}{k^2}$ are uniformly bounded, in order to conclude that the series is convergent. This can be seen as follows:

$$\begin{aligned} \sum_{k=1}^n \frac{1}{k^2} &= 1 + \sum_{k=2}^n \frac{1}{k^2} \\ &\leq 1 + \sum_{k=2}^n \frac{1}{k(k-1)} \\ &= 1 + \sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k} \right) \\ &= 1 + \left(1 - \frac{1}{n} \right) \\ &< 2, \quad \forall \quad n \geq 2. \end{aligned}$$